Seminar 8 (25.11.2025)

1. Prove that for $(\forall)\ x, y \in \mathbb{R}^m$ the following hold:

a) $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$

b) $\langle x, y\rangle = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2)$

a) Euclidian scalar product: $\langle x, y\rangle = x \cdot y = x_1y_1 + \dots + x_my_m$

Euclidean norm $\|x\| = \sqrt{\langle x, x\rangle} = \sqrt{x_1^2 + x_2^2 + \dots + x_m^2}$

↳ length of a vector with m components.

Method 1. (using Euclidean scalar product)

$\|x+y\|^2 = (x_1+y_1)^2 + (x_2+y_2)^2 + \dots + (x_m+y_m)^2$ (1)

$\|x-y\|^2 = (x_1-y_1)^2 + (x_2-y_2)^2 + \dots + (x_m-y_m)^2$. (2)

(1): $x_1^2 + 2x_1y_1 + y_1^2 + x_2^2 + 2x_2y_2 + y_2^2 + \dots + x_m^2 + 2y_mx_m + y_m^2$

(2): $x_1^2 - 2x_1y_1 + y_1^2 + x_2^2 - 2x_2y_2 + y_2^2 + \dots + x_m^2 - 2x_2y_2 + y_2^2$

(1) + (2): $2x_1^2 + 2y_1^2 + 2x_2^2 + 2y_2^2 + \dots + 2x_m^2 + 2y_m^2$

$= 2(x_1^2 + y_1^2 + x_2^2 + y_2^2 + \dots + x_m^2 + y_m^2)$

$= 2(x_1^2 + \dots + x_m^2) + 2(y_1^2 + \dots + y_m^2)$

$= 2 \cdot \|x\|^2 + 2\|y\|^2$ a) ✓

$\|x\|^2 = x_1^2 + \dots + x_m^2$

(1) − (2): $4(x_1y_1) + 4x_2y_2 + \dots + 4x_my_m$

$= 4 \cdot \langle x, y\rangle$. b) ✓.

$\Rightarrow \langle x, y\rangle = \frac{1}{4}((\|x+y\|)^2 - \|x-y\|^2)$

Method 2. (random scalar prod.) $\longrightarrow$ Consider a scalar prod. $\langle \cdot, \cdot \rangle$

$\|x\| = \sqrt{\langle x, x\rangle}$.

(*) $\|x+y\|^2 = \left(\sqrt{\langle x+y, x+y\rangle}\right)^2 = \langle x+y, x+y\rangle = \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2$.

(**) $\|x-y\|^2 = \sqrt{\langle x-y, x-y\rangle}^2 = \langle x-y, x-y\rangle =$

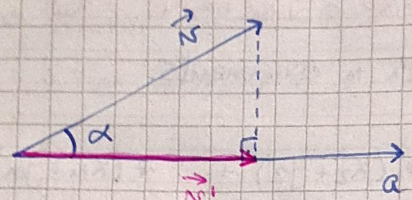
$= \langle x, x\rangle - \langle x, y\rangle - \langle y, x\rangle + \langle y, y\rangle = \|x\|^2 - 2\langle x, y\rangle + \|y\|^2$.

$(*) + (**) \Rightarrow \|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$

$(*) - (**) \Rightarrow \|x+y\|^2 - \|x-y\|^2 = 4\langle x, y\rangle.$

$$\Rightarrow \langle x, y\rangle = \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2\right).$$

(2.) Find the orthogonal projection of a vector $v \in \mathbb{R}^2$ onto a vector a ...



$$\cos\alpha = \frac{\overset{(\text{cat. al})}{\|v'\|}}{\underset{(\text{ip})}{\|v\|}} \Rightarrow \|v'\| = \cos\alpha \cdot \|v\|$$

$$\boxed{\langle a, v\rangle = \|a\| \cdot \|v\| \cdot \cos\alpha}, \cancel{\cos\alpha}$$

$$\Rightarrow \cos\alpha = \frac{\langle a, v\rangle}{\|a\| \cdot \|v\|}$$

$$\Rightarrow \|v'\| = \frac{\langle a, v\rangle}{\|a\| \cdot \cancel{\|v\|}} \cdot \cancel{\|v\|}$$

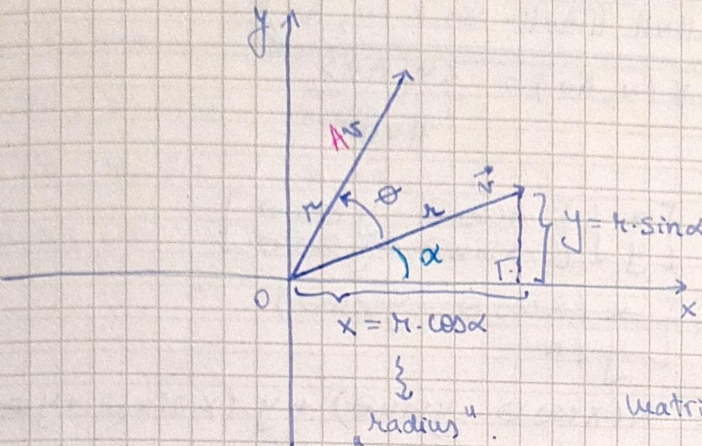
$$\Rightarrow \boxed{\|v'\| = \frac{\langle a, v\rangle}{\|a\|}}$$

$$v' = \underbrace{\frac{1}{\|a\|} \cdot a}_{\text{unit vector}} \cdot \|v'\|$$

$$v' = \frac{a}{\|a\|} \cdot \frac{\langle a, v\rangle}{\|a\|} = \frac{\langle a, v\rangle}{\|a\|^2} \cdot a = \frac{\langle a, v\rangle}{\langle a, a\rangle} \cdot a$$

this is the projection.

③. Show that $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is a rotation matrix with angle θ in $\mathbb{R}^2$.



matrix × vector = vector.

$\cancel{v = (x, y)}$

$\cancel{Av =}$

$v = \begin{bmatrix} x \\ y \end{bmatrix}$

$$Av = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}$$

Method 1

$$v = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} r\cos\alpha \\ r\sin\alpha \end{bmatrix}$$

$$A \cdot v = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \cdot \begin{bmatrix} r\cos\alpha \\ r\sin\alpha \end{bmatrix} = \begin{bmatrix} r\cos\theta\cos\alpha - r\sin\theta\sin\alpha \\ r\sin\theta\cos\alpha + r\cos\theta\sin\alpha \end{bmatrix}$$

$$= \begin{bmatrix} r\cdot\cos(\theta+\alpha) \\ r\cdot\sin(\theta+\alpha) \end{bmatrix}$$

$\boxed{\text{! } v, Av \left[\begin{array}{l} \text{have the same length } r. \\ \text{have a} \neq \text{angle} \end{array}\right] \text{ this is a ROTATION.}}$

$\sin(\theta+\alpha)$
$\cos\cdots$

Method 2

$\boxed{\begin{array}{l} \text{1) Show that } \|v\| = \|Av\|. \\ \text{2) Show that } \measuredangle(v, Av) = \theta. \end{array}}$ $\Big\} \Rightarrow$ ROTATION.

$$\|Av\| = \sqrt{(x\cos\theta - y\sin\theta)^2 + (x\sin\theta + y\cos\theta)^2}$$

$$= \sqrt{x^2\cos^2\theta \cancel{- 2xy\cos\theta\sin\theta} + y^2\sin^2\theta + x^2\sin^2\theta \cancel{+ 2xy\sin\theta\cos\theta} + y^2\cos^2\theta}$$

$$= \sqrt{x^2(\sin^2\theta + \cos^2\theta) + y^2(\sin^2\theta + \cos^2\theta)}$$

$$= \sqrt{x^2 + y^2} = \|v\|$$

$\Rightarrow \|Av\| = \|v\|. \quad (1).$

$$\downarrow$$

$$\boxed{\langle v, Av\rangle = \|v\| \cdot \|Av\| \cdot \cos \measuredangle (v, Av)}$$

$$\Rightarrow \quad \cos \measuredangle (v, Av) = \frac{\langle v, Av\rangle}{\|v\|^2}$$

$$\langle v, Av\rangle = \begin{bmatrix} x \\ y \end{bmatrix} \cdot \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}$$

$$\cancel{= \begin{cases} x(x\cos\theta - y\sin\theta) \\ y(x \end{cases}}$$

$$= x(x\cos\theta - y\sin\theta) + y(x\sin\theta + y\cos\theta)$$

$$= x^2\cos\theta - \cancel{xy\sin\theta} + \cancel{xy\sin\theta} + y^2\cos\theta$$

$$= \cos\theta \, (x^2 + y^2)$$

$$= \cos\theta \cdot \|v\|^2.$$

$$\Rightarrow \langle v, Av\rangle = \|v\| \cdot \|v\| \cdot \cos\theta$$

$$\Downarrow$$

$$\measuredangle (v, Av) = \theta.$$

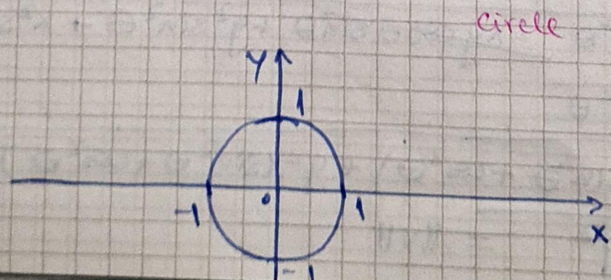
4. Consider the p-norm $\|x\|_p := (|x_1|^p + \dots + |x_m|^p)^{\frac{1}{p}}$, $x \in \mathbb{R}^m$, $p \geq 1$,
and $\|x\|_\infty := \max \{ |x_1|, |x_2|, \dots, |x_m| \}$.
Represent the unit ball in $\mathbb{R}^2$ for $p = \begin{cases} 1, \\ 2, \\ \infty, \end{cases}$ ....

$p = 2 \Rightarrow \|x\|_2 = \sqrt{|x_1|^2 + |x_2|^2} = \sqrt{x_1^2 + x_2^2}$, $x = (x_1, x_2)$

the euclidean norm.

Unit ball $= \{ x = (x_1, x_2) \in \mathbb{R}^2 \mid \|x\|_2 = 1 \}$

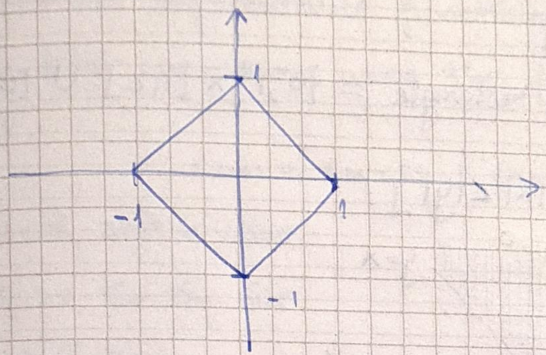
$$= \{ (x_1, x_2) \in \mathbb{R}^2 \mid \underbrace{x_1^2 + x_2^2 = 1}_{\text{circle}} \}$$



$p=1 \Rightarrow \|x\| = |x_1|+|x_2|$.

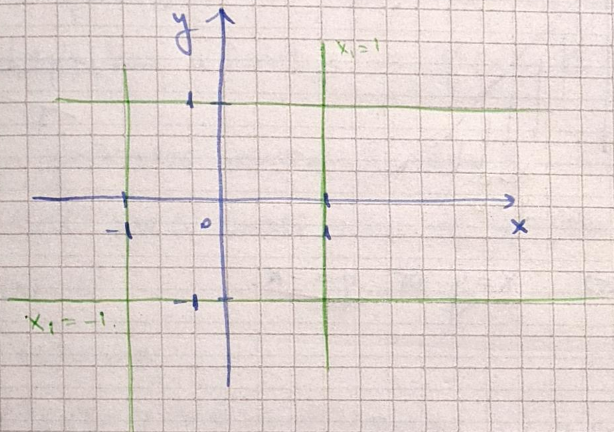
<u>Unit ball</u> $= \{(x_1,x_2)\in\mathbb{R}^2 \mid |x_1|+|x_2| = 1\}$.

- $x_1, x_2 > 0$: $x_1 + x_2 = 1$
- $x_1 > 0,\ x_2 < 0$: $x_1 - x_2 = 1$
- $x_1 < 0,\ x_2 > 0$: $-x_1 + x_2 = 1$
- $x_1, x_2 < 0$: $-x_1 - x_2 = 1$



$p=\infty \Rightarrow \|x\|_\infty = \max\{|x_1|, |x_2|\}$

<u>Unit ball</u> $= \{(x_1,x_2)\in\mathbb{R}^2 \mid \max\{|x_1|,|x_2|\} = 1\}$

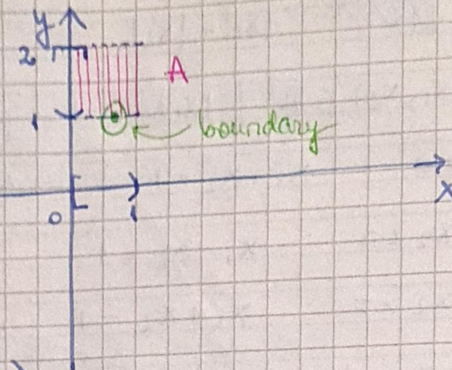


$\text{int}\, A = \{x\in\mathbb{R}^m \mid \exists V\in\mathcal{V}(x),\ V\subseteq A\}$.

$\text{cl}\, A = \{x\in\mathbb{R}^m \mid \forall V\in\mathcal{V}(x),\ V\cap A\neq\emptyset\}$

$\text{bd}\, A = \{x\in\mathbb{R}^m \mid \forall V\in\mathcal{V}(x),\ V\cap A\neq\emptyset \text{ and } V\cap A^c\neq\emptyset\}$

5) a) $A = [0,1) \times (1,2]$

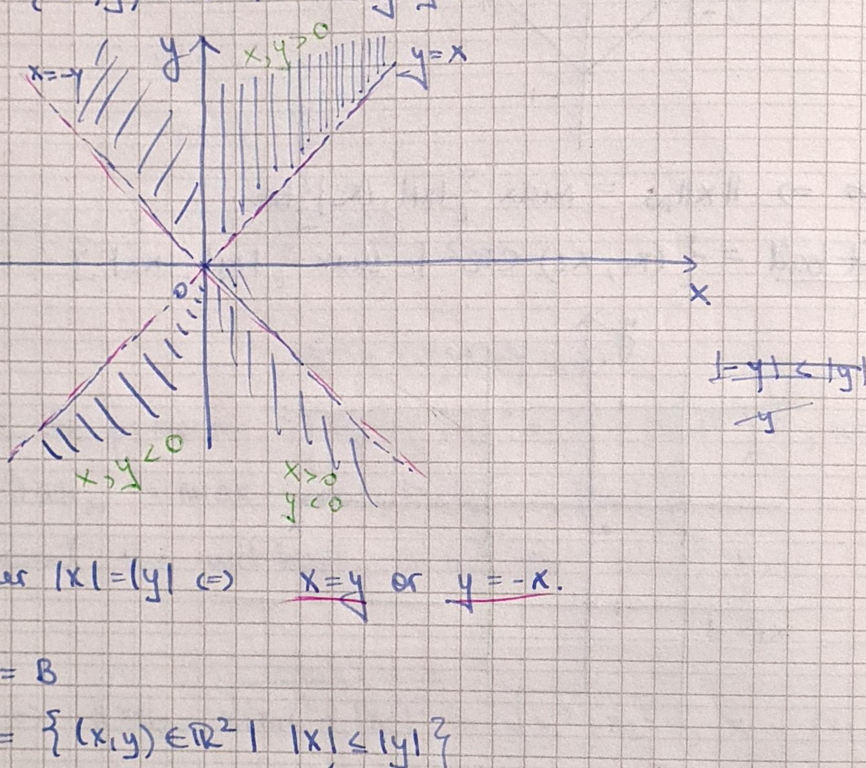


$\text{int}\, A = (0,1) \times (1,2)$.

$\text{cl}\, A = [0,1] \times [1,2]$

$\text{bd}\, A =$ sides of the rectangle $= \{0,1\} \times [1,2] \cup [0,1] \times \{1,2\}$.

b) $B = \{ (x,y) \in \mathbb{R}^2 \mid |x| < |y| \}$.



Consider $|x| = |y| \iff x = y$ or $y = -x$.

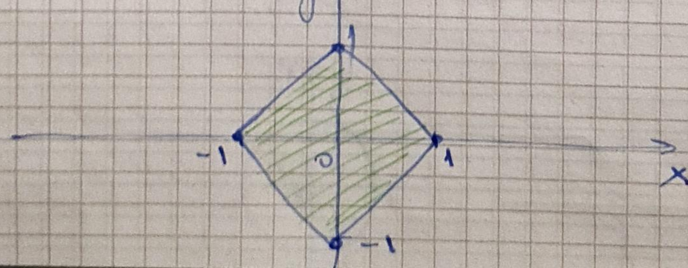
$\text{int}\, B = B$

$\text{cl}\, B = \{ (x,y) \in \mathbb{R}^2 \mid |x| \leq |y| \}$

$\text{bd}\, B = \{ (x,y) \in \mathbb{R}^2 \mid |x| = |y| \}$.

c) $C = \{ (x,y) \in \mathbb{R}^2 \mid |x| + |y| < 1 \}$.

Consider $|x| + |y| = 1$.



$\text{int } C = C$

$\text{cl } C = \{(x,y) \in \mathbb{R}^2 \mid |x| + |y| \leq 1\}$.

$\text{bd } C = \{(x,y) \in \mathbb{R}^2 \mid |x| + |y| = 1\}$.

↳ boundary of the square.

The unit ball for that

p-norm when $p = 1$.